

Denormalization

Some consequences of normalization

Data redundancy is reduced or eliminated.

- Computed data eliminated.

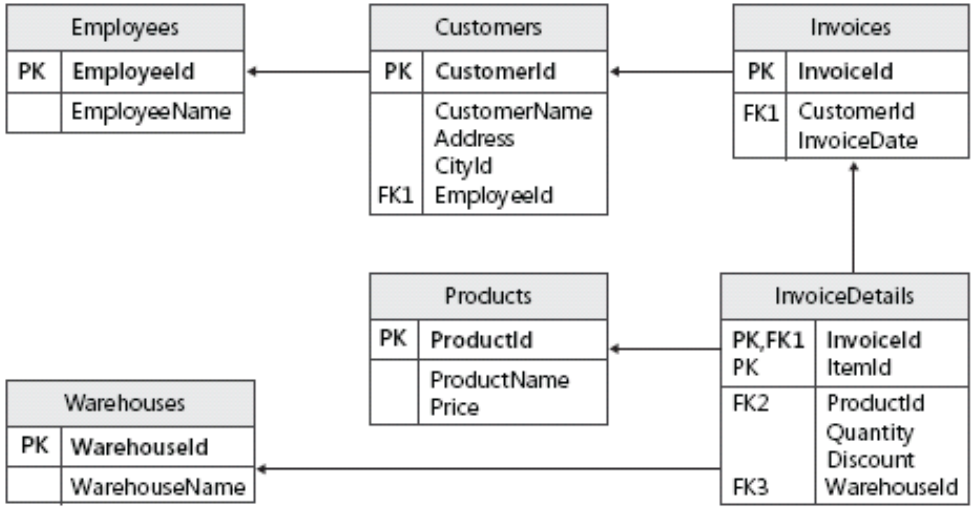
Relations broken into smaller, related tables.

- Must join if need original relation.

Denormalization

Deliberately reintroducing some redundancy, so that we can access data faster.





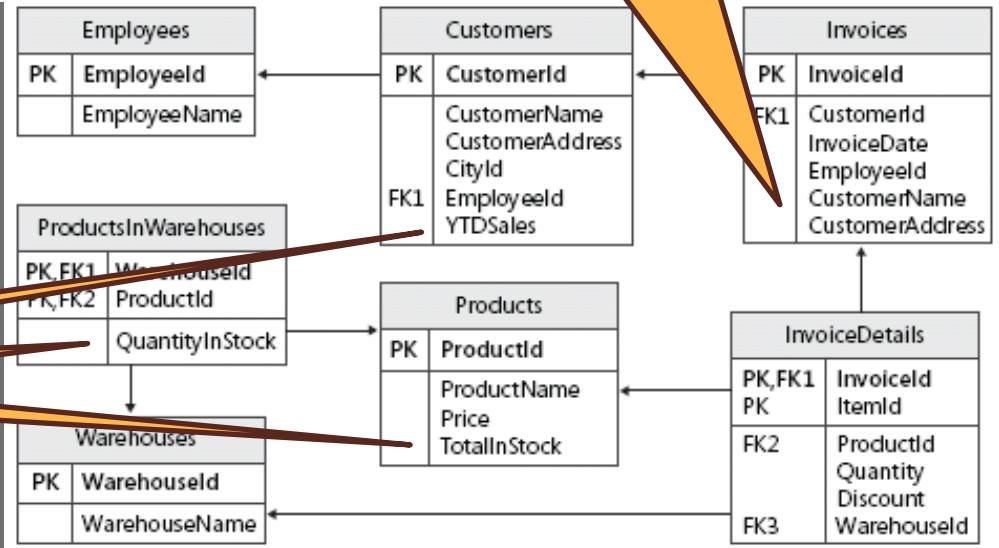
Techniques

Why? Avoid joins

Add duplicate fields

Add computed fields

Why? Avoid computation



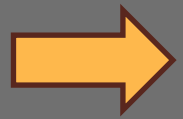
Technique: Join tables

Why?
Avoid joins

prod_sku	prod_id	color_id	prod_descr	color
1	1	1	Mountain bike 100	red
2	1	2	Mountain bike 100	black
3	2	1	Road bike 200	red
4	2	2	Road bike 200	black
5	2	3	Road bike 200	silver
6	3	1	Hybrid bike 50	red
7	3	3	Hybrid bike 50	silver
8	3	4	Hybrid bike 50	blue

Technique: Split table

client	region	…
1	U.S.	…
2	Asia	…
3	U.S.	…
4	Asia	…
5	Africa	…



Regions
Subclasses
Frequent/rare

client	region	…
1	U.S.	…
3	U.S.	…

client	region	…
2	Asia	…
4	Asia	…

client	region	…
5	Africa	…

Why?

- Less data per table
- Distribute data near users

If writing/updating redundant data

Data consistency requires work.

Is full consistency required at all times?

Maintain consistency via

- API functions
- Triggers
- Application code

Fix/reconcile consistency via

- Background process
- System maintenance